

SOCIOL 723: Problem Set 2

Assigned: Friday, September 18 • **Due:** Friday, October 2, 11:59PM

Your name:

Instructions. Submit a single PDF rendered from a Quarto or RMarkdown source, together with the source file. This problem set is about what you will do in research: reading a published table, fitting the same kind of model on real data, and writing up what it says. No question asks for a derivation; every computation uses the functions from Labs 3 and 4, and the outputs reproduced in the first section are read, not re-run. Where a question asks for sentences, write them as you would in a paper. You may use AI assistance, but you are responsible for every line you submit. Use `set.seed(723)` wherever there is randomness. The GSS extract is `Data/gss_earnings.rds`; the school data are `nlme::MathAchieve` and `nlme::MathAchSchool`, merged as in Lab 4.

Maximum Likelihood, in Practice

The three outputs below come from the GSS extract. Output 1 shows the last lines of `summary()` for the logit `ba ~ pareduc + wordsum + female + black`, followed by `logLik()` of the same model. Outputs A and B show what `glm()` prints when asked to trace its iterations, `control = glm.control(trace = TRUE)`: the deviance after each step. Output A is the same model as Output 1. Output B fits `ba ~ pareduc + perfect`, where `perfect` is 1 for respondents with at least 18 years of schooling who hold a degree and 0 otherwise, and shows its coefficient table.

Output 1

```
Null deviance: 4751.9  on 3508  degrees of freedom
Residual deviance: 3958.8  on 3504  degrees of freedom
AIC: 3968.8
```

```
Number of Fisher Scoring iterations: 4
```

```
'log Lik.' -1979.402 (df=5)
```

Output A

```
Deviance = 3971.34 Iterations - 1
Deviance = 3958.874 Iterations - 2
Deviance = 3958.804 Iterations - 3
Deviance = 3958.804 Iterations - 4
```

Output B

```
Deviance = 3548.226 Iterations - 1
Deviance = 3460.042 Iterations - 2
Deviance = 3435.541 Iterations - 3
Deviance = 3427.344 Iterations - 4
Deviance = 3424.467 Iterations - 5
Deviance = 3423.429 Iterations - 6
Deviance = 3423.05 Iterations - 7
Deviance = 3422.911 Iterations - 8
Deviance = 3422.86 Iterations - 9
Deviance = 3422.841 Iterations - 10
Deviance = 3422.834 Iterations - 11
Deviance = 3422.832 Iterations - 12
Deviance = 3422.831 Iterations - 13
Deviance = 3422.83 Iterations - 14
Deviance = 3422.83 Iterations - 15
Deviance = 3422.83 Iterations - 16
Deviance = 3422.83 Iterations - 17
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.899	0.207	-18.811	0.000
pareduc	0.232	0.015	15.742	0.000
perfect	19.320	271.710	0.071	0.943

converged: TRUE

1. **The idea, in one curve.** Let θ be the share of adults with a bachelor's degree, $k \leftarrow \text{sum}(\text{gss\$ba})$ successes in $n \leftarrow \text{nrow}(\text{gss})$ draws. For $\text{theta} \leftarrow \text{seq}(0.2, 0.5, \text{by} = 0.001)$, compute $\text{dbinom}(k, n, \text{theta}, \text{log} = \text{TRUE})$ and plot it against theta . This is the log-likelihood: the same $\text{dbinom}()$ you would use for a probability, read with the data fixed and θ moving. Mark the highest point and confirm it is $\text{mean}(\text{gss\$ba})$. Two sentences: what does “maximum likelihood estimate” mean, and what does the width of the peak tell you?
2. **Reading the likelihood lines (Output 1).** The model has five parameters. (a) Show how the AIC line follows from the log-likelihood line. (b) “Number of Fisher Scoring iterations: 4”: what happened four times, and why did it stop? (c) The null deviance is for the intercept-only model and the residual deviance for this one. What quantity from Week 3 is their difference, what does it test, and on how many degrees of freedom?
3. **Converged, or not? (Outputs A and B).** (a) In which output has the search settled? Say what in the trace tells you, one sentence each. (b) R reports `converged: TRUE` for Output B as well. Look at the coefficient on `perfect` and its standard error. Would you

report that row? What value is the coefficient trying to reach, and why can no finite number satisfy the likelihood here? (c) Which coefficients in Output B did the algorithm pin down, and which did it not? In two sentences, say what you would do before reporting anything from this model.

Reading a Published Logit: Killewald (2016)

Killewald (2016) asks whether a couple's economic circumstances predict divorce and whether the answer has changed across marriage cohorts. The data are couple-years from the Panel Study of Income Dynamics, 1968 to 2013: one row per couple per year, with the outcome equal to 1 in the year a couple divorces. Table 3 fits a logit separately to marriages begun in 1974 or earlier and in 1975 or later; Figure 1 converts the main coefficients into predicted probabilities of divorce in the next year. The rows of Table 3 used below, and Figure 1, are reproduced on the next page.

Reference: Killewald, Alexandra. 2016. "Money, Work, and Marital Stability: Assessing Change in the Gendered Determinants of Divorce." *American Sociological Review* 81(4): 696–719.

1. **One coefficient, three scales.** Take the husband's full-time employment in the later cohort, $-.293$. Write one sentence on each scale: log-odds, odds (use the printed odds ratio), and probability (from Figure 1: 2.5% when he is employed full-time, 3.3% when he is not). Then compute the odds ratio implied by those two probabilities and compare it with the printed .746. Should the two agree exactly? Answer in one or two sentences, using the note under Figure 1 on how the probabilities were computed.
2. **A coefficient that is not significant.** In the later cohort, the wife's full-time employment has coefficient .069 and standard error .088; Figure 1 shows 2.6% when she is employed full-time and 2.5% when she is not. Write the sentence that belongs in a results section. In the earlier cohort the same coefficient is .267 with standard error .153, and Figure 1 shows 1.3% against 1.0%. Killewald writes that the association declined across cohorts but that the decline is not statistically significant. Compute the difference between the two coefficients and its standard error, $\sqrt{.153^2 + .088^2}$, and use them to explain in two sentences what supports each half of her claim.
3. **Units matter.** The wife's proportion of housework runs from 0 to 1, and its coefficient in the earlier cohort is -1.117 , printed odds ratio .327. What change in housework does that odds ratio describe? Figure 1 plots a smaller change, from 50% to 75%. Start from the figure's 1.5% at 50%, multiply the odds by $e^{-1.117 \times .25}$, and convert back to a probability; you should land on the figure's 1.1%. Then one sentence on why you cannot recover the odds ratio precisely by working backwards from the printed 1.5% and 1.1%, and one sentence on why a coefficient's size means nothing until you know the units of its variable.

	Married 1974 or earlier			Married 1975 or later		
	$\hat{\beta}$	(SE)	OR	$\hat{\beta}$	(SE)	OR
Wife employed full-time	.267	(.153)	1.306	.069	(.088)	1.071
Husband employed full-time	-.097	(.208)	.908	-.293**	(.106)	.746
Wife's proportion of housework	-1.117*	(.463)	.327	.050	(.233)	1.051
Constant	-7.476***	(.884)		-3.525***	(.393)	
Couple-years	32,853			33,470		

Killewald (2016), Table 3, excerpt. Logit coefficients, standard errors, and odds ratios as printed. * $p < .05$; ** $p < .01$; *** $p < .001$. Controls include marital duration, religion, race, home ownership, children, both spouses' education, region, and year of marriage.

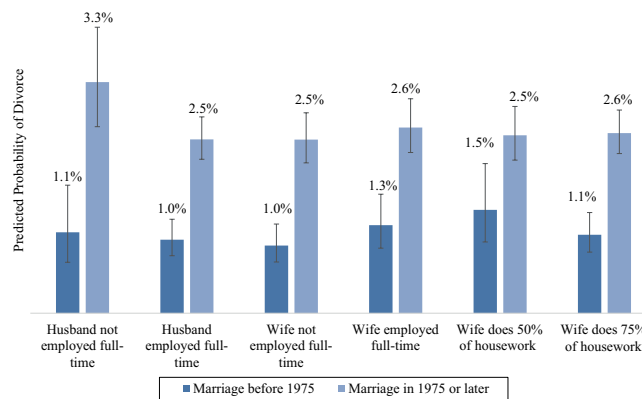


Figure 1. Predicted Probabilities of Divorce in the Next Year
Note: Probabilities are predicted based on the logistic regression coefficients in Table 3 for a respondent with all other covariates set to the cohort-specific means. Error bars are 95% confidence intervals, estimated by calculating the bounds of the confidence interval for the predicted log odds (non-bootstrapped) and then transforming those bounds to predicted probabilities.

Killewald (2016), Figure 1, reproduced for classroom use.

Your Own Logit, Reported Like a Paper

Fit the logit from lecture on the GSS extract:

```
m1 <- glm(ba ~ pareduc + wordsum + female + black, data = gss, family = binomial)
```

1. **Table and figure.** Report the coefficient table with odds ratios added, laid out like Killewald's Table 3. Then make your own Figure 1: with `avg_predictions`, compute the average predicted probability of a degree when everyone's parents have 8, 12, 16, and 20 years of education, with 95% intervals, and plot the four bars with error bars. Two sentences: why is the gain from 12 to 16 years not the same as the gain from 16 to 20, when the coefficient is one number?
2. **An interaction, read correctly.** Does parents' education matter more for women? Fit `ba ~ pareduc * female + wordsum + black`. Report the product coefficient with its z , then the AME of `pareduc` separately for men and women and the difference with

its standard error. Answer the question in a short paragraph and say why the product coefficient alone could not have answered it.

3. **A block of controls.** Add the region dummies (`region`) to `m1`. Report the likelihood-ratio test, AIC, and BIC for the two models. In a short paragraph, say what each is measuring, why with $n = 3,509$ the LR test rejects almost any block, and which model you would report and why.
4. **The results paragraph.** Write the paragraph (120 to 180 words) that would appear in the Results section of a paper reporting this model. Use predicted probabilities or AMEs, not odds ratios; give the reader magnitudes and uncertainty; do not mention the log-odds scale.

Counts

Fit the Poisson regression for the number of children:

```
m_pois <- glm(childs ~ educ + female + black + exper + evermar, data = gss, family = poisson)
```

1. **Reading it, and which standard errors.** Interpret $e^{\hat{\beta}}$ for `educ` and for `evermar`, one sentence each, in the form you would use in a paper. Then run `AER::dispersiontest(m_pois)` and report the standard error of `evermar` from the Poisson fit, from a quasi-Poisson fit, and from `sandwich::vcovHC`. Two sentences: what does the test say about the Poisson's variance assumption, and which standard error would you report as your default and why?
2. **Zeros, and a model with two parts.** Report the observed share of childless respondents and the share the Poisson predicts (average `dpois(0, fitted(m_pois))`). Fit the hurdle model with `pscl::hurdle(..., dist = "poisson")` and the same covariates in both parts. Make one table with the log-likelihood, number of parameters, AIC, and BIC of the Poisson and the hurdle. Then, in a paragraph written for a reader, say what `evermar` does to the chance of *any* child, what it does to the number of children *among parents*, and what the single Poisson coefficient was averaging over.

Observations in Groups

Merge `nlme::MathAchieve` with `nlme::MathAchSchool` as in Lab 4, and convert `School` to a plain factor with `factor(as.character(School))`, as the lab does: 7,185 students in 160 schools, math achievement, student SES, and school sector.

1. **From clustered standard errors to a random intercept.** Fit `lm(MathAch ~ SES + Sector)` and report the coefficient on `Sector` with its plain standard error and with its standard error clustered on `School`. In a short paragraph, explain to a first-year student why the plain standard error is wrong for this variable: sector is a property of the school, so it varies across 160 schools, not across 7,185 students. Then fit `lmer(MathAch ~ SES +`

`Sector + (1 | School), REML = FALSE)` and report the two coefficients with standard errors, $\hat{\tau}^2$, and $\hat{\sigma}^2$. Say in one sentence each what the two variances measure, compute the intraclass correlation, and close with two sentences on what the random-intercept model gives you that the clustered standard error did not.

2. **Three estimates of one slope, and three questions.** Fit the fixed-effects regression `lm(MathAch ~ School + SES + Sector)`, with the school dummies listed *before Sector*. One coefficient comes back NA: which one, and why? What would you have concluded had you listed `Sector` before `School` and read off the number R printed? Then compare the SES slope under pooled OLS, the random intercept, and fixed effects in one three-column table, and explain in a paragraph where the random-intercept slope sits relative to the other two and why, in terms of the variation each estimate uses. Finally, for each research question below, say in two sentences which of *OLS with clustered standard errors*, *fixed effects*, or a *random intercept* you would use, naming the feature of the question that decides it.
 - (a) How much of earnings inequality in a country is between workplaces rather than within them, and is that share rising?
 - (b) Does a worker's wage rise after their own firm signs a union contract, using several years of data on the same workers and firms?
 - (c) Do students learn more in Catholic schools, using these 160 schools, where sector is a property of the school?